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GCAMUSAJobs: An R package for employment projections based on GCAM-USA power sector outcomes

Di Sheng [1], **Brian O'Neill** [1], **Stephanie Waldhoff** [1], and **Matthew Binsted** [1]

1 Joint Global Change Research Institute, Pacific Northwest National Laboratory, College Park, MD, USA

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Software

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Summary

The `GCAMUSAJobs` R package was developed to post-process electric power projections from GCAM-USA, enabling the estimation of future power sector jobs at the state-level by generation technology and job type. `GCAMUSAJobs` extends GCAM-USA functionality by (1) estimating the capacity levels of different activities – operational capacity, capacity addition, and retirement; and (2) calculating jobs associated with production activities, including those in operation and maintenance (O&M), construction, and decommissioning.

Statement of need

The development of `GCAMUSAJobs` was driven by the need to assess the distributional labor impacts of energy system transition (Hanson, 2023; Mayfield et al., 2023; Raimi, 2021; Xie et al., 2023). While gross employment (Mayfield et al., 2023) and power sector employment (Xie et al., 2023) are expected to grow into the future, over time under both business as usual and decarbonization, Xie et al. (2023) find insignificant differences in power sector jobs between the two scenarios. Other research has also suggested that fossil fuel-intensive states may experience slower job growth or job losses (Hanson, 2023; Mayfield et al., 2023).

Currently, GCAM-USA does not calculate power sector jobs. `GCAMUSAJobs` addresses this gap by providing projected direct power sector jobs based on GCAM-USA output, enhancing the functionality of GCAM-USA for labor impact analysis.

Workflow

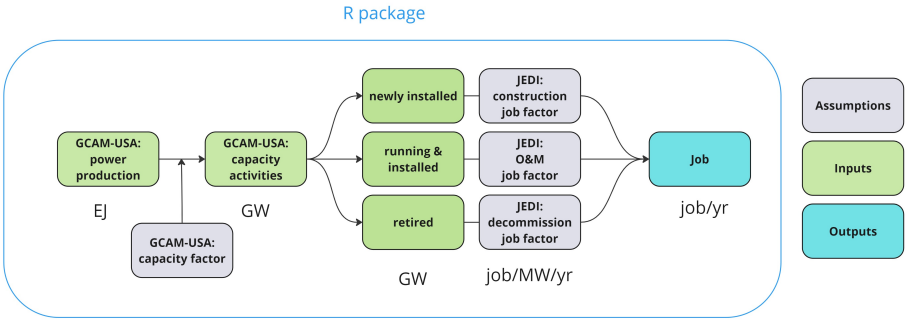

Figure 1: Figure. 1. Package workflow.

`GCAMUSAJobs` utilizes GCAM-USA annual electricity generation outputs to estimate underlying capacity levels based on assumptions about capacity factors and calculate associated power sector jobs based on employment factors (Fig. 1). The employment factor represents the average number of jobs created per unit of power production activity (e.g., jobs per gigawatt). This method is widely used in the relevant literature (Mayfield et al., 2023; Rutovitz et al., 2015). `GCAMUSAJobs` adopts employment factors from NREL's Jobs & Economic Development Impacts (JEDI) model (https://www.nrel.gov/analysis/jedi/models.html), which has been broadly used in the literature (Jacobson et al., 2017; Rutovitz et al., 2015; Xie et al., 2023).

Key functions

`GCAMUSAJobs::GCAM_EJ` queries power generation data (in exajoules, EJ) from the GCAM-USA output database for a single scenario, disaggregating generation from existing plants, newly added plants, and the generation lost from recently retired plants. The output is provided annually, broken down by state and fuel technology. Building on this, `GCAMUSAJobs::GCAM_GW`, taking the output from `GCAMUSAJobs::GCAM_EJ`, calculates the average annual capacity levels (in gigawatts, GW) by state and fuel technology for different activities, including operation, addition, and retirement. It supports both the "Total" and "Net" methods. The "Total" method allows for both new capacity addition and premature retirement of a given technology to happen in the same period, while the "Net" method assumes that only the difference between these two activities would occur. It therefore offsets the addition by premature retirement, providing adjusted capacity levels by activities. `GCAMUSAJobs::GCAM_JOB` then utilizes the output from `GCAMUSAJobs::GCAM_GW` to estimate the average annual job estimates, broken down by fuel type and job type, including construction (both on-site and construction-related), operations & maintenance, and decommissioning. Users can select between the "Total" or "Net" method, with "Total" used as the default. `GCAMUSAJobs` also provides a list of functions to visualize the employment factor assumptions, capacity, and job outcomes.

`GCAMUSAJobs::GCAM_EJ` is compatible with both the GCAM-USA output database as well as a project data file queried using the R package `rgcam`. Please refer to the package vignette for additional examples and visualizations.

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Implementation

For demonstration purposes, we use `GCAMUSAJobs` to post-process the outcome from GCAM v7.1 for a standard reference scenario, estimating the direct job, aggregated over states, associated with U.S. power generation (Fig. 2).

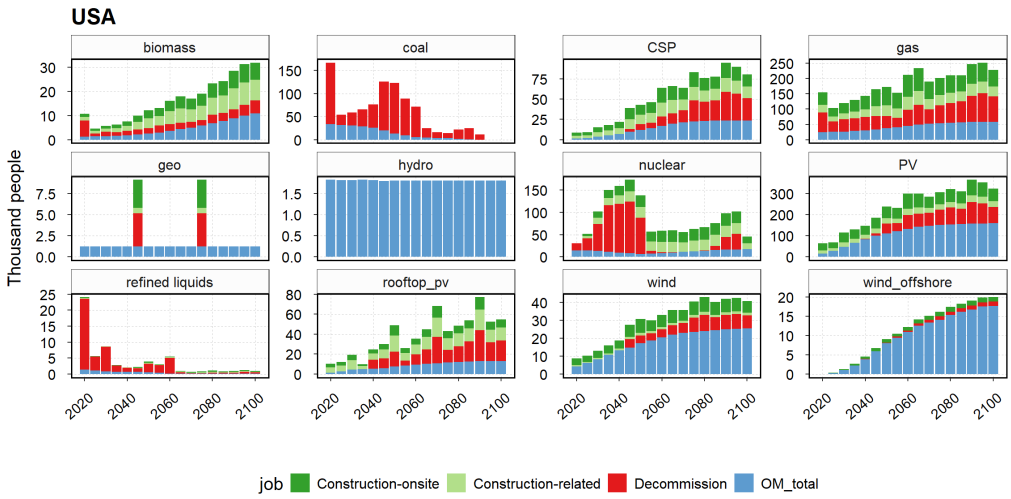

Figure 2: Figure. 2. Annual average power sector jobs by fuel and job types over a 5-year model period. Note that y-axes have different scales.

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References

Hanson, G. H. (2023). *Local labor market impacts of the energy transition: Prospects and policies*. National Bureau of Economic Research.

Jacobson, M. Z., Delucchi, M. A., Bauer, Z. A., Goodman, S. C., Chapman, W. E., Cameron, M. A., Bozonnat, C., Chobadi, L., Clonts, H. A., Enevoldsen, P., & others. (2017). 100% clean and renewable wind, water, and sunlight all-sector energy roadmaps for 139 countries of the world. *Joule*, *1*(1), 108–121.

Mayfield, E., Jenkins, J., Larson, E., & Greig, C. (2023). Labor pathways to achieve net-zero emissions in the united states by mid-century. *Energy Policy*, *177*, 113516.

Raimi, D. (2021). Mapping county-level exposure and vulnerability to the US energy transition. *Resources for the Future Working Paper*, 21–36.

Rutovitz, J., Dominish, E., & Downes, J. (2015). *Calculating global energy sector jobs: 2015 methodology*.

Xie, J. J., Martin, M., Rogelj, J., & Staffell, I. (2023). Distributional labour challenges and opportunities for decarbonizing the US power system. *Nature Climate Change*, *13*(11),

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